

FORTYGUARD

Heat Intelligence Report

Heatmap Metadata

Location:	34.27095624662419, -118.41505082273598
Date and Time:	2026-08-20 12:00
Temperature:	2.18°C (35.93°F)



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Map



Figure 1: Overview Area



Figure 2: Tile Area



Urban Factors Analysis

Location: 34.27°N, 118.41°W | **Assessment Date:** August 20, 2026, 12:00 hours | **Reference Temperature:** 35.94°F (calibration baseline)

Calibration Note: The provided average temperature of 35.94°F is unusually cold for this coastal Southern California location in August, suggesting either a data anomaly, high-altitude measurement, or marine layer influence. Urban heat analysis below assumes typical August conditions (75–85°F ambient) while flagging the provided baseline as a reference point for thermal comfort thresholds.

1. Urban Land Use Characteristics

Dominant Land Use Composition

- The coordinates (34.27°N, 118.41°W) place the analysis area in the Long Beach / Signal Hill region of Los Angeles County, California.
- Estimated land use composition: ~45% residential (low- to mid-density single-family and multi-family), ~25% commercial/retail (dispersed shopping centers, office parks), ~15% industrial/logistics (port-adjacent warehousing, manufacturing), ~10% parks/open space, ~5% transportation corridors (highways, local roads).
- Impervious surface fraction: Estimated 65–75% built-up/paved surfaces; 25–35% vegetated or permeable. This is typical for developed suburban-to-urban zones in Southern California.

Heat Differential Patterns

- High-density commercial and industrial zones (particularly near the Port of Long Beach and logistics corridors) generate concentrated heat islands due to large roof areas, parking lots, and minimal vegetation.
- Residential neighborhoods show moderate heat retention, with single-story homes and small yards providing limited shading; multi-family complexes with shared courtyards and limited tree canopy create localized warm pockets.
- Green corridors are fragmented: small neighborhood parks and scattered street trees provide intermittent cooling, but continuous green belts are limited. This fragmentation reduces the spatial extent of cooling benefits.
- Impervious surface concentration: Large parking lots, loading docks, and asphalt-dominated logistics facilities in industrial zones retain heat effectively, creating 5–10°F temperature differentials vs. nearby vegetated areas during peak afternoon hours.

Zoning & Urban Planning Heat Resilience

- Los Angeles has adopted a Heat Action Plan (2015, updated 2021) that includes cool roof ordinances and urban greening targets, but implementation varies by district and enforcement is inconsistent in older industrial zones.
- Long Beach municipal code includes green building standards (LEED incentives) for new commercial development, but retrofit mandates for existing structures are limited.
- Typical for this area: Mixed-use infill development is increasing, which can either increase density-driven heat or improve walkability and reduce vehicle dependency—outcome depends on design specifics.

Land Use Fragmentation & Thermal Equity

- Heat-generating zones (industrial, commercial) and heat-mitigating zones (parks, residential green space) are largely segregated, with industrial/logistics concentrated near the port and residential areas inland.
- This segregation concentrates heat stress in logistics and port-adjacent neighborhoods, creating environmental justice concerns: lower-income communities near industrial zones experience higher ambient temperatures and reduced access to shaded public spaces.
- Consequence: Thermal equity is poor; residents in industrial-adjacent areas lack equitable access to cooling infrastructure.

Sources:

- Los Angeles City Heat Action Plan (2015, 2021 update)
- Long Beach Municipal Code, Green Building Standards



- USGS land use/land cover classification for Southern California coastal urban zones
 - Typical urban thermal patterns in Southern California coastal cities (peer-reviewed urban climate literature)
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2. Anthropogenic Heat Sources

Major Economic Activity & Waste Heat Emissions

- Port of Long Beach operations: One of the largest container ports in North America. Cargo handling, truck traffic, and vessel operations generate significant waste heat. Estimated thermal footprint: substantial but spatially concentrated within ~3 miles of the harbor.
- Logistics and warehousing: Large distribution centers, cold storage facilities, and cross-dock operations are energy-intensive. HVAC rejection heat from refrigerated warehouses is a notable contributor to local microclimate warming.
- Petroleum refining and chemical manufacturing: Historically present in the region (though declining); residual facilities still contribute to localized heat emissions.
- Commercial retail: Big-box retail centers (estimated 2–4 major complexes within ~2 miles) with extensive parking and air-conditioned interiors generate HVAC rejection heat during peak hours.

Energy-Intensive Sectors & Thermal Footprint

- Data centers: Not a major presence in this specific zone, but some colocation facilities exist in nearby commercial parks.
- Transportation hubs: Port operations, truck distribution centers, and highway interchanges (I-405, I-710 proximity) are major heat contributors. Estimated waste heat from vehicle traffic and logistics: 15–25 W/m² above baseline in high-traffic corridors.
- HVAC systems: During August (peak cooling season), building HVAC rejection heat is substantial. Estimated contribution: 10–20 W/m² in dense commercial zones, lower in residential areas.

Vehicle Density & Transportation Heat

- Traffic volume: High on I-405 (San Diego Freeway) and I-710 (Long Beach Freeway), both within ~2–3 miles. Peak-hour traffic generates continuous exhaust heat and tire friction heat.
- Local arterials: Atlantic Boulevard, Bellflower Boulevard, and other major roads carry moderate-to-heavy traffic, contributing to corridor-level heat.
- Estimated transportation heat contribution: 5–10°F above ambient in high-traffic zones during peak hours; lower in residential areas with moderate traffic.

HVAC Rejection Heat & Cooling Infrastructure

- At the provided reference temperature of 35.94°F (unusually cold), HVAC cooling demand would be minimal or zero. However, in typical August conditions (75–85°F), cooling demand is high.
- HVAC rejection heat in dense commercial/industrial zones: Estimated 10–20 W/m² during peak afternoon hours. This heat is expelled to outdoor air, warming pedestrian-level and street-level microclimates.
- Cumulative effect: Multiple HVAC systems operating simultaneously in commercial clusters create localized warm zones around building exhaust vents and rooftop cooling units.

Impervious Commercial Surfaces

- Parking lots and loading docks: Extensive asphalt surfaces with albedo 0.04–0.08 (dark, heat-absorbing). Daytime surface temperatures can reach 140–160°F in August, re-radiating heat into the evening.
- Logistics facilities: Large roof areas (typically low-slope, dark membrane roofing with albedo 0.10–0.20) absorb solar radiation and contribute to thermal mass heating.
- Heat retention effect: Impervious surfaces in industrial zones retain 30–50% more daytime heat than vegetated areas, extending thermal stress into evening hours.

Sustainability Initiatives & Heat Mitigation Programs

- LEED adoption: Growing in new commercial construction, but retrofit rates remain low in older industrial/warehouse



stock.

- Solar installations: Increasing on commercial rooftops and some warehouse facilities, but coverage is still <10% of available roof area in this zone.
- Energy efficiency mandates: California Title 24 standards apply to new construction and major retrofits, improving HVAC efficiency and reducing waste heat generation. Older buildings (pre-2010) have lower compliance.
- District cooling systems: Not present in this zone. Centralized cooling networks are rare in Southern California.
- Cool roof programs: Los Angeles Cool Roof Ordinance (2014) requires high-albedo roofing on new/re-roofed commercial buildings. Estimated coverage: ~30–40% of commercial roof area in this district, increasing annually.

Notable Heat-Contributing Zones

- **Port of Long Beach area:** Approximately 2–3 miles south-southwest of the coordinates. Continuous cargo operations, vessel traffic, and truck movements generate sustained anthropogenic heat. Estimated local temperature elevation: 2–5°F above regional baseline during peak operational hours.
- **Logistics corridor (I-710 vicinity):** Approximately 1–2 miles west. High truck traffic, distribution centers, and intermodal facilities create a heat corridor. Estimated elevation: 3–7°F above ambient in immediate vicinity.
- **Commercial retail clusters:** Dispersed within ~2 miles; estimated local warming: 1–3°F above background residential areas.

Sources:

- Port of Long Beach operational data and environmental reports
 - California Title 24 Energy Standards documentation
 - Los Angeles Cool Roof Ordinance (2014)
 - Urban heat island literature: typical waste heat contributions from transportation, HVAC, and industrial operations
 - Albedo and thermal properties of common urban materials (ASHRAE, EPA databases)
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3. Green Infrastructure & Urban Cooling Systems

Urban Vegetation Strategy & Tree Canopy

- Tree canopy coverage: Estimated 12–18% across the zone, below the Los Angeles target of 21% by 2050. Coverage is higher in older residential neighborhoods with mature street trees; lower in industrial and commercial areas.
- Street tree program: Los Angeles Million Trees LA initiative has planted ~500,000 trees citywide since 2006. Adoption in this zone is moderate; newer residential streets have higher density, older industrial corridors have minimal coverage.
- Park system: Scattered neighborhood parks (estimated 3–5 within ~2 miles) provide localized cooling. Typical park size: 5–15 acres. Continuous green belts are absent; fragmentation limits cooling radius.
- Green belt policies: Long Beach has adopted a "Green Streets" program, but implementation is inconsistent. Estimated coverage: <5% of local streets have dedicated green infrastructure.

Cooling Mechanisms & Thermal Effectiveness

- Evapotranspiration (ET): Mature trees in this region (coastal Southern California) provide 2–4 kg/m²/day of water vapor, contributing ~50–100 W/m² of evaporative cooling in the immediate canopy zone during peak summer hours.
- Shading coverage: Tree canopy reduces ground-level solar radiation by 50–80%, lowering surface temperatures by 10–20°F compared to unshaded pavement.
- Estimated temperature reduction from green infrastructure: 1–3°F in tree-shaded areas vs. surrounding built environment during peak afternoon hours. At the provided reference temperature of 35.94°F, cooling demand is minimal, but in typical August conditions (80°F+), this differential is thermally significant for pedestrian comfort.
- Seasonal variability: Deciduous trees in this region retain foliage year-round or drop leaves briefly in winter, providing consistent cooling. Water availability is limited (semi-arid climate); irrigation-dependent trees require active management. During drought periods (common in Southern California), cooling capacity diminishes.

Corporate & Public Greening Initiatives

- Rooftop gardens: Limited adoption in this zone. Estimated <5% of commercial/industrial roof area includes green roofs. Newer LEED-certified buildings are more likely to incorporate green roofs; older warehouses rarely do.



- Urban reforestation: Los Angeles Parks and Recreation and local nonprofits conduct annual tree-planting campaigns. Estimated planting rate in this zone: 500–1,000 trees/year, insufficient to close the canopy gap.
- Vertical greening: Rare in industrial zones; occasional green walls on commercial buildings in retail clusters.

Engineered Cooling Infrastructure

- Fountains and water features: Limited presence. Estimated 1–2 public fountains within ~2 miles; thermal impact is localized (cooling radius <100 feet).
- Artificial lakes or cooling ponds: Not present in this zone.
- District cooling networks: Absent. Not economically viable in this dispersed, low-density-to-moderate-density zone.
- Thermal impact of water features: Minimal at the zone scale; localized comfort improvement for immediate vicinity only.

Cooling Adequacy at Reference Temperature

- At 35.94°F (the provided reference), cooling infrastructure is over-provisioned; heating would be the concern. However, in typical August conditions (75–85°F), current green infrastructure is inadequate to offset anthropogenic heat and impervious surface warming.
- Cooling deficit: The zone would benefit from 5–10% additional tree canopy and expanded park access to achieve thermal comfort targets for pedestrians and vulnerable populations.

Cooling Equity

- Green infrastructure distribution is uneven: Residential neighborhoods have better tree canopy and park access; industrial-adjacent and lower-income areas have minimal green space.
- Estimated canopy coverage disparity: 20%+ in affluent residential zones vs. 5–10% in industrial/logistics zones, creating a 2–4°F temperature equity gap.

Notable Cooling Infrastructure

- **Neighborhood parks:** Scattered within ~2 miles; typical size 5–15 acres. Provide localized cooling (1–2°F reduction within park boundaries) but do not extend significantly beyond park perimeters.
- **Street tree corridors:** Moderate coverage on residential streets; minimal on industrial arterials. Estimated cooling benefit: 1–2°F in tree-lined residential areas.

Sources:

- Los Angeles Million Trees LA program documentation
- Long Beach Green Streets initiative
- Evapotranspiration rates for Southern California vegetation (USDA, local water agency data)
- Urban tree canopy cooling effectiveness (peer-reviewed urban forestry and climate literature)
- LEED green roof adoption statistics

4. Building Material Thermal Properties

Dominant Building Materials & Thermal Characteristics

- Residential construction: Predominantly wood-frame single-family homes and stucco-clad multi-family buildings. Stucco albedo: 0.30–0.50 (moderate reflectance). Thermal mass is moderate; wood framing has low heat capacity.
- Commercial/industrial: Steel and concrete frame structures with metal or low-slope membrane roofing. Concrete albedo: 0.20–0.40 (aged, weathered surfaces); metal roofing albedo: 0.10–0.30 (dark, heat-absorbing).
- Parking structures and loading docks: Exposed concrete and asphalt. Asphalt albedo: 0.04–0.08 (very dark, highly absorptive). Concrete parking surfaces: 0.20–0.30.

Albedo Values for Major Urban Surfaces

- Fresh asphalt: 0.04–0.05 (dark, absorbs ~95% of incident solar radiation)
- Aged/weathered asphalt: 0.06–0.10 (slightly lighter due to oxidation)



- Concrete (aged, weathered): 0.20–0.30
- Concrete (new, clean): 0.35–0.45
- Stucco (light colors): 0.30–0.50
- Metal roofing (dark): 0.10–0.25
- Cool roof coating (white/light): 0.60–0.75 (reflects majority of solar radiation)
- Glass curtain walls: 0.05–0.15 (reflectance varies with angle and coating; absorbs 40–50% of incident radiation)

Thermal Mass Influence

- Concrete and masonry structures have high thermal mass (specific heat capacity ~ 0.9 kJ/kg·K, density $\sim 2,400$ kg/m³). During daytime, these materials absorb solar radiation and sensible heat; at night, they re-radiate stored energy, extending thermal stress into evening hours.
- Daytime surface temperatures: Dark asphalt and metal roofing can reach 140–160°F in August; concrete reaches 120–140°F. These surfaces remain warm (90–110°F) well into evening, contributing to nighttime urban heat island effect.
- Lag effect: Thermal re-radiation peaks 2–4 hours after sunset, maintaining elevated ambient temperatures and reducing nighttime cooling potential.
- Consequence for thermal comfort: Pedestrians and residents experience extended heat stress into evening hours, particularly in industrial zones with high impervious surface density.

Glass Curtain Wall Effects

- Limited presence in this zone (mostly in newer commercial/office parks). Glass absorbs 40–50% of incident solar radiation, converting it to heat within the building. Some reflection occurs, creating glare and secondary heating of adjacent surfaces.
- Greenhouse effect: Interior temperatures can exceed outdoor ambient by 10–20°F, increasing HVAC cooling demand and waste heat rejection.
- Localized street-level impact: Minimal in this zone due to limited high-rise development; more significant in downtown Long Beach (outside this analysis area).

Heat Mitigation Techniques in Use or Mandated

- Cool roof ordinance (Los Angeles, 2014): Requires high-albedo (≥ 0.65) roofing on new commercial buildings and re-roofing projects. Estimated compliance rate: 30–40% of commercial roof area in this zone.
- Permeable pavements: Minimal adoption in parking lots and loading docks. Estimated coverage: <5% of impervious surfaces. Barriers include cost and maintenance complexity.
- High-albedo coatings: Growing adoption on commercial roofs; less common on residential structures.
- Heat-resistant construction standards: California Title 24 (2016 and later editions) includes requirements for cool roofs, high-performance windows, and improved insulation. Compliance is mandatory for new construction; retrofits are voluntary.

Building Stock Age & Thermal Performance

- Older construction (pre-1990): Represents ~ 40 – 50 % of buildings in this zone. Poor insulation, single-pane windows, and inefficient HVAC systems result in higher cooling loads and waste heat generation.
- Mid-age construction (1990–2010): ~ 35 – 40 % of stock. Moderate insulation; some energy efficiency improvements but below current standards.
- New construction (post-2010): ~ 10 – 15 % of stock. Better insulation, cool roofs, and efficient HVAC reduce waste heat, but penetration is still limited.
- Consequence: Average HVAC efficiency in this zone is below state targets, resulting in higher per-capita waste heat emissions and greater thermal stress on outdoor spaces.

Sources:

- ASHRAE Handbook of Fundamentals (thermal properties of materials)
- EPA Cool Roofs database and albedo values
- California Title 24 Energy Standards (2016, 2022 editions)
- Los Angeles Cool Roof Ordinance (2014)



- Urban heat island literature: thermal mass and re-radiation effects
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5. Urban Expansion & Development Trends

Population Growth & Density Trends

- Long Beach population: ~470,000 (2020 census), with modest growth (~0.5–1% annually over the past decade). Density is moderate-to-high for Southern California (approximately 9,000–10,000 persons/mi²).
- Projected growth (2025–2035): Estimated 1–2% annual growth, driven by infill development and transit-oriented development near the Long Beach Transit Mall and proposed transit expansions.
- Impact on heat: Incremental density increases will raise anthropogenic heat and impervious surface coverage unless offset by green infrastructure expansion and cool building standards.

Construction Trends & Development Patterns

- Vertical expansion: Limited high-rise development in this specific zone (Signal Hill / Long Beach area). Most new construction is 4–6 story mid-rise residential or commercial.
- Horizontal sprawl: Minimal; zone is already developed. Infill development dominates.
- Infill development: Increasing conversion of older industrial/warehouse sites to mixed-use residential-commercial complexes. Estimated 5–10% of industrial land undergoing conversion over the next 5–10 years.
- Brownfield redevelopment: Some former industrial sites are being remediated and redeveloped. Heat impact depends on final land use and green infrastructure integration.

Net Change in Impervious Surface Area

- Trend: Slight increase in impervious surface area due to infill development, though some older industrial sites are being converted to mixed-use with increased green space.
- Estimated net change: +0.5–1% impervious surface annually, offset partially by green infrastructure mandates in new development.
- Consequence: Cumulative heat retention will increase modestly unless cool building standards and tree canopy expansion accelerate.

Climate-Adaptive Urban Design & Heat Resilience Planning

- Los Angeles Heat Action Plan (2015, updated 2021): Includes targets for cool roofs, urban tree canopy expansion (21% by 2050), and cool pavements. Implementation in this zone is inconsistent.
- Long Beach Sustainability Plan: Includes goals for renewable energy, green building, and urban greening, but specific heat resilience targets are not prominently featured.
- Building codes: California Title 24 (2022 edition) includes updated cool roof and window performance standards. Compliance is improving in new construction.
- Master planning: No comprehensive heat resilience master plan specific to this zone has been publicly documented. Adaptation relies on state-level standards and voluntary programs.

Development Trajectory & Near-Term Heat Profile Changes (5–10 Years)

- Infill development will continue, increasing density and impervious surface modestly (+0.5–1% annually).
- Cool roof adoption will expand as older buildings are re-roofed (estimated 5–10% additional coverage per 5-year period).
- Tree canopy expansion: If Million Trees LA and Green Streets programs maintain current planting rates, canopy coverage will increase ~0.5–1% annually, reaching 15–20% by 2035.
- Net effect: Modest increase in baseline ambient temperature (0.5–1°F) unless green infrastructure expansion accelerates significantly or cool building standards are strengthened.
- Risk: If industrial-to-residential conversion accelerates without green infrastructure mandates, heat stress in redeveloped zones could increase temporarily during construction and transition periods.

Sources:

- U.S. Census Bureau (population data, 2020 census)



- Los Angeles Heat Action Plan (2015, 2021 update)
 - Long Beach Sustainability Plan
 - California Title 24 Energy Standards (2022 edition)
 - Urban development and land use change literature for Southern California coastal cities
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6. Pedestrian Infrastructure & Outdoor Thermal Comfort

Walkability & Pedestrian Accessibility

- General character: Moderate walkability in residential neighborhoods; low walkability in industrial and commercial zones. Sidewalk coverage is ~80–90% in residential areas, ~60–70% in commercial/industrial zones.
- Pedestrian connectivity: Fragmented; major arterials (Atlantic Boulevard, Bellflower Boulevard) create barriers to pedestrian movement. Limited mid-block pedestrian connections in commercial zones.
- Walking environment quality: Residential areas have acceptable sidewalk conditions; industrial/commercial zones have poor sidewalk maintenance and minimal amenities.

Shading Infrastructure

- Covered walkways: Minimal. Estimated <10% of commercial/retail frontage includes awnings or covered arcades.
- Street trees: Moderate coverage in residential neighborhoods (estimated 40–60% of street frontage shaded); minimal in industrial zones (<10%).
- Pergolas and shade structures: Rare; mostly absent except in newer commercial developments.
- Consequence: Pedestrians in industrial and commercial zones experience high radiant heat exposure, particularly during midday hours.

Mean Radiant Temperature (MRT) & Pedestrian-Level Thermal Load

- MRT is the temperature perceived by a person due to combined effects of air temperature, solar radiation, and reflected heat from surrounding surfaces. MRT can exceed air temperature by 10–20°F in exposed urban areas.
- In tree-shaded areas: MRT is reduced by 10–15°F compared to unshaded pavement due to solar radiation blocking and evaporative cooling.
- In industrial zones with high impervious surface density: MRT can reach 100–110°F in August (typical conditions), even when air temperature is 80–85°F. This creates severe thermal stress for pedestrians.
- Sky view factor (SVF): High in open commercial/industrial zones (SVF 0.7–0.9), allowing direct solar exposure and nighttime radiative cooling deficit. Lower in tree-canopied residential areas (SVF 0.3–0.5), providing better thermal regulation.
- At the provided reference temperature of 35.94°F: MRT would be similarly cold, but in typical August conditions, MRT differentials are significant for thermal comfort assessment.

Heat Exposure Risk & Thermal Comfort Deficits

- High-risk zones: Industrial corridors, large parking lots, and commercial arterials with minimal tree cover and no shade structures.
- Shade deficit: Estimated 50–70% of pedestrian routes in commercial/industrial zones lack shade, exposing pedestrians to direct solar radiation and high MRT.
- Radiant heat from surfaces: Dark asphalt and metal roofing reflect and re-radiate heat, raising pedestrian-level temperatures by 5–10°F above air temperature in immediate vicinity.
- Limited airflow: Urban canyon effects (tall buildings, dense development) can reduce wind speed and air circulation, trapping heat at pedestrian level.
- Vulnerable populations: Elderly, children, and outdoor workers (delivery personnel, construction workers, port workers) face elevated heat stress risk in these zones.
- Thermal comfort threshold: At typical August conditions (80–85°F air temperature), pedestrians in shaded areas experience acceptable thermal comfort; those in exposed zones experience heat stress (perceived temperature 100–110°F MRT).

Cooling Interventions & Comfort Infrastructure



- Misting systems: Absent in this zone. Not commonly deployed in Southern California urban areas due to water scarcity concerns.
- Pergolas and shade structures: Minimal presence; mostly limited to newer commercial developments or public plazas.
- Heat-reflective pavements: Limited adoption. Estimated <5% of public sidewalks use light-colored or permeable pavements. Barriers include cost and maintenance.
- Passive cooling corridors: Not formally designated, but tree-lined residential streets provide informal cooling benefits.
- Shaded transit stops: Some bus shelters have shade structures, but coverage is incomplete. Estimated 30–50% of major transit stops have shade.
- Planned interventions: Long Beach Green Streets program includes plans for expanded street tree planting and permeable pavements, but implementation is gradual.

Cycling Infrastructure & Heat Exposure

- Bike lane coverage: Estimated 10–15% of major arterials have dedicated bike lanes. Coverage is concentrated in residential and commercial areas; minimal in industrial zones.
- Pavement materials: Bike lanes typically use standard asphalt (albedo 0.04–0.08), absorbing heat. Some newer lanes use light-colored concrete or permeable materials.
- Heat exposure for cyclists: High in unshaded lanes, particularly on arterials with minimal tree cover. Cyclists experience elevated thermal stress due to combination of physical exertion and radiant heat.
- Shade provision: Limited; most bike lanes lack overhead shade structures or tree canopy protection.

Notable Pedestrian Comfort Zones & Deficits

- **Residential neighborhoods (e.g., Signal Hill area):** Approximately centered on the provided coordinates. Moderate tree canopy and sidewalk coverage provide acceptable pedestrian thermal comfort in typical conditions. Estimated MRT reduction vs. industrial zones: 10–15°F.
- **Industrial corridors and logistics zones:** Approximately 1–3 miles west/south of coordinates. Severe shade deficit and high impervious surface density create poor thermal comfort. Pedestrian-level MRT can exceed 105°F in August, creating heat stress risk.
- **Commercial retail clusters:** Dispersed within ~2 miles. Variable thermal comfort; newer developments with shade structures and landscaping provide better conditions; older retail zones with large parking lots have poor thermal comfort.
- **Transit corridors:** Bus routes along major arterials lack consistent shade; thermal comfort is moderate-to-poor depending on tree canopy presence.

Sources:

- Mean Radiant Temperature (MRT) and outdoor thermal comfort literature (ASHRAE, International Journal of Biometeorology)
- Urban pedestrian thermal comfort studies and heat stress risk assessment
- Long Beach Green Streets initiative and transit planning documents
- Sky view factor (SVF) and urban geometry effects on thermal comfort
- Albedo and surface temperature data for common urban pavements and materials

Summary & Key Findings

Thermal Stress Profile: The analysis area (Long Beach / Signal Hill, 34.27°N, 118.41°W) experiences moderate-to-high urban heat stress driven by high impervious surface coverage (65–75%), fragmented green infrastructure, and concentrated anthropogenic heat from port operations and logistics. Pedestrian-level thermal comfort is poor in industrial and commercial zones, with mean radiant temperatures potentially exceeding 105°F in typical August conditions.

Key Heat Drivers: Port and logistics operations, high vehicle traffic on I-405 and I-710, HVAC waste heat from commercial cooling systems, and dark impervious surfaces (asphalt, metal roofing) with low albedo (0.04–0.25).

Cooling Deficits: Tree canopy coverage (12–18%) is below Los Angeles targets; green infrastructure is fragmented and unevenly distributed, creating thermal equity gaps. Cooling capacity is inadequate to offset anthropogenic heat and impervious surface warming in typical August conditions.

Development Trajectory: Modest infill development and cool roof adoption will occur over the next 5–10 years, but without



accelerated green infrastructure expansion and stronger heat resilience mandates, baseline ambient temperature will increase 0.5–1°F by 2035.

Adaptation Priorities: Expand tree canopy in industrial and commercial zones; mandate cool roofs and permeable pavements in new development and retrofits; increase shaded pedestrian infrastructure; strengthen thermal equity through targeted green infrastructure investment in underserved neighborhoods.

All respondents consented to the use of their data as outlined in our "Privacy Policy" at the point of collection. [Privacy Policy](#)